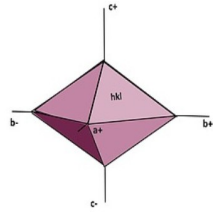
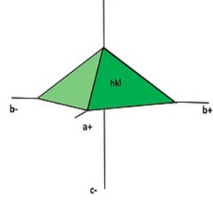
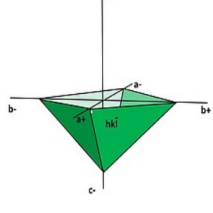
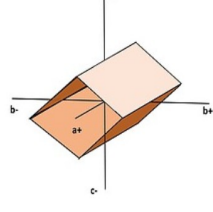
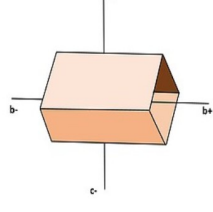
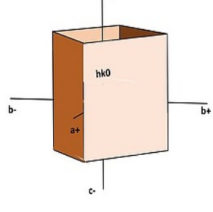
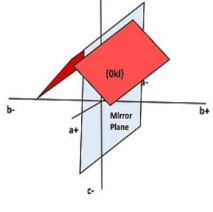
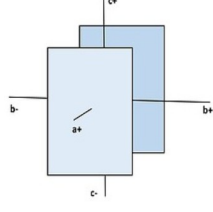
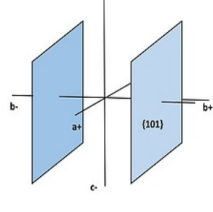
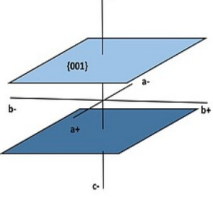
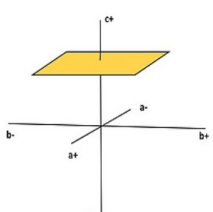


Systèmes Cristallins: Exemples de Géométrie

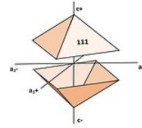
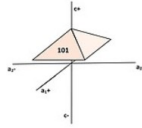
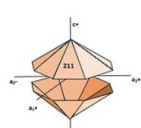
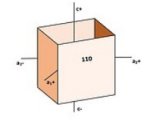
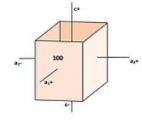
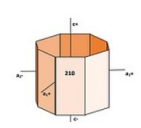
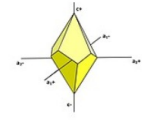
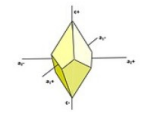
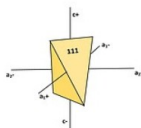
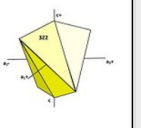
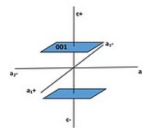
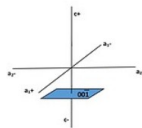
Système Orthorhombique ($\alpha = \beta = \gamma = 90^\circ$)

Formes fréquentes

Dipyramid & Pyramids	 Orthorhombic dipyramid	 Orthorhombic Pyramid $\{hkl\}$; Positive	 Orthorhombic Pyramid $\{hkl\}$; Negative	
Prisms & Domes	 Orthorhombic Prism $\{0kl\}$; 1st Order	 Orthorhombic Prism $\{h0l\}$; 2nd Order	 Orthorhombic Prism $\{hk0\}$; 3rd Order	 Orthorhombic Dome $\{0kl\}$; 1st Order
Pinacoids & Pedions	 Orthorhombic Pinacoid $\{100\}$	 Orthorhombic Pinacoid $\{010\}$	 Orthorhombic Pinacoid $\{001\}$	 Pedion

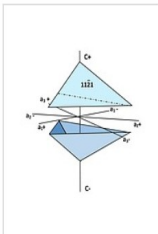
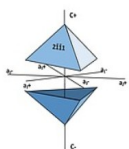
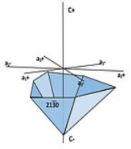
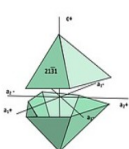
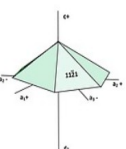
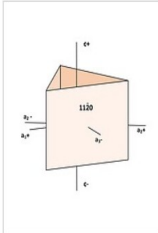
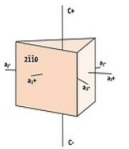
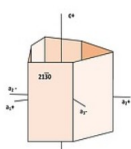
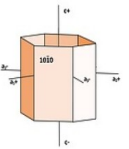
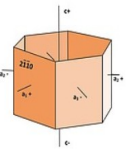
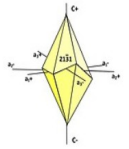
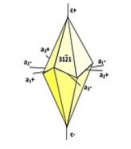
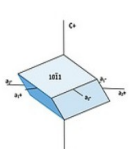
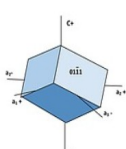
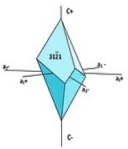
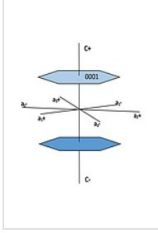
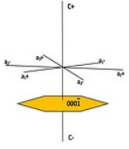
Système Tétragonal ($a = b$, $\alpha = \beta = \gamma = 90^\circ$)

Formes fréquentes

Pyramids & Dipyramids	 Tetragonal Dipyramid 1st Order	 Tetragonal Pyramid 2nd Order Positive	 Ditetragonal Dipyramid	
Prisms & Diprisms	 Tetragonal Prism 1st Order	 Tetragonal Prism 2nd Order	 Ditetragonal Prism	
Trapezohedrons; Disphenoid; & Scalenohedron	 Tetragonal Trapezohedron Left	 Tetragonal Trapezohedron, Right	 Tetragonal Disphenoid	 Tetragonal Scalenohedron Positive
Pinacoid & Pedion	 Tetragonal Pinacoid	 Tetragonal Pedion, basal		

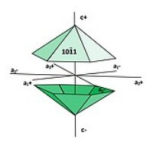
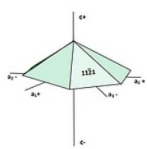
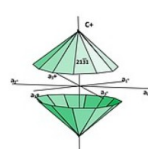
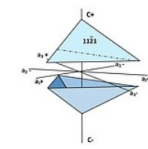
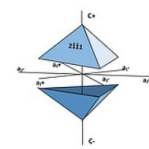
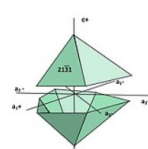
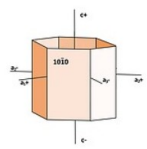
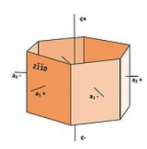
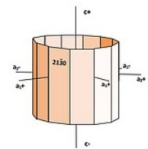
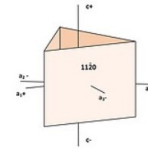
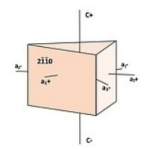
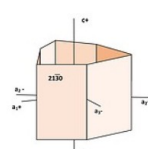
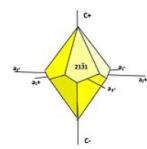
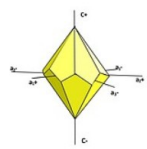
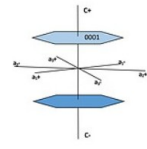
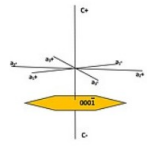
Système Trigonal ($a = b$, $\alpha = \beta = \gamma = 90^\circ$)

Formes fréquentes

Pyramids and Dipyramids	 Trigonal Dipyramid: 1st Order	 Trigonal Dipyramid 2nd Order	 Ditrigonal Pyramid 1st Order	 Ditrigonal Dipyramid	 Hexagonal Pyramid: 2nd Order		
Prisms & Diprisms	 Trigonal Prism: 1st Order	 Trigonal Prism: 2nd Order	 Ditrigonal Prism	 Hexagonal Prism 1st Order	 Hexagonal Prism 2nd Order		
Scalohedrons	Rhombohedrons & Trapezohedrons		 Trigonal Scalohedron Positive	 Trigonal Scalohedron Negative	 Rhombohedron: Positive	 Rhombohedron: Negative	 Trigonal Trapezohedron: Left Positive
Pinacoids & Pedions	 Hexagonal Pinacoid (001)	 Hexagonal Pedion					

Système Hexagonal ($a = b$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$)

Formes fréquentes

Pyramids & Dipyramids						
Prisms & Diprisms						
Trapezohedron						
Pinacoid & Pedion						

Système Monoclinique ($\alpha = \gamma = 90^\circ$)

Formes fréquentes

Prisms & Domes				
	{hkl}; 4th Order Prism Positive	{0kl}; 1st Order Prism	{hk0}; 3rd Order Prism	3rd Order Dome
Pinacoids				
	{h0l}; 2nd Order Pinacoid Positive	{100}; a Pinacoid	{010}; b Pinacoid	{001}; c Pinacoid; Basal Pinacoid

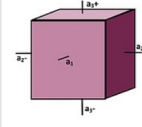
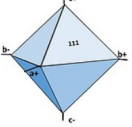
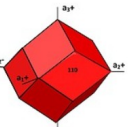
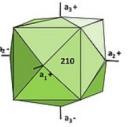
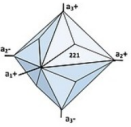
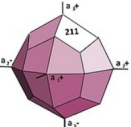
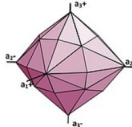
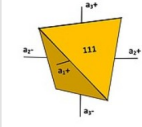
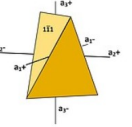
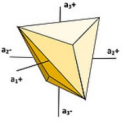
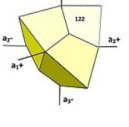
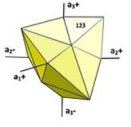
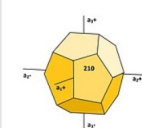
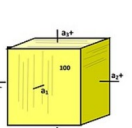
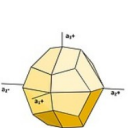
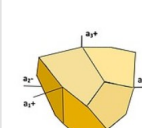
Système Triclinique (-)

Formes fréquentes

Triclinic Pinacoid $\{h00\}$ $\{0k0\}$ $\{00l\}$; (3 of 3)				
	Front $\{100\}$	Side $\{010\}$	Basal $\{001\}$	
Triclinic Pinacoid $\{0kl\}$ 1st Order; (2 of 2)				
	Positive $\{011\}$	Negative $\{0\bar{1}0\}$		
Triclinic Pinacoid $\{h0l\}$ 2nd Order; (2 of 2)				
	Positive $\{101\}$	Negative $\{10\bar{1}\}$		
Triclinic Pinacoid $\{hk0\}$ 3rd Order; (2 of 2)				
	Positive $\{110\}$	Negative $\{1\bar{1}0\}$		
Triclinic Pinacoid $\{hkl\}$ 4th Order; (4 of 4)				
	Positive Right $\{111\}$	Negative Right $\{111\}$	Negative Left $\{111\}$	Positive Left $\{111\}$
Triclinic Pedions; (4 of 26)				
	$\{100\}$	$\{111\}$	$\{110\}$	$\{111\}$

Système Isométrique ($a = b = c$, $\alpha = \beta = \gamma = 90^\circ$)

Formes fréquentes

Hexoctahedral Class							
	Isometric Hexahedron	Isometric Octahedron	Isometric Dodecahedron	Isometric Tetrahexahedron	Isometric Trisoctahedron	Isometric Trapezohedron	Isometric Hexoctahedron
Hextetrahedral Class							
	Isometric Tetrahedron: Positive	Isometric Tetrahedron: Negative	Isometric Tristetrahedron: Positive	Isometric Deltoid Dodecahedron: Positive	Isometric: Hextetrahedron Positive		
Diploidal Class							
	Isomeric Pyritohedron Positive	Isometric Striated Pyrite	Isometric Diploid				
Tetartoidal Class							
	Isometric Tetartohedron Right Positive						

Familles Cristallines, Systèmes Cristallins et Réseaux de Bravais

Famille Cristalline	Système Cristallin	Système de Réseau	Groupes d'Espace	Réseaux de Bravais	Notation Hermann-Mauguin
Isométrique	Isométrique	Isométrique	36	P, I, F	m-3m, 4/m-3-2/m, 432, -43m, 23
Tétragonal	Tétragonal	Tétragonal	68	P, I	4/mmm, 422, 4mm, -42m, 42m, 4/m, 4
Hexagonal	Hexagonal	Hexagonal	27	P	6/mmm, 622, 6mm, -6m2, 6/m, -6, 6
Hexagonal	Trigonal	Hexagonal	25	P	-3m, 3m, 32, -3, 3
Hexagonal	Trigonal	Rhomboédrique	7	R	-3m, 3m, 32, -3, 3
Orthorhombique	Orthorhombique	Orthorhombique	59	P, C, I, F	mmm, 222, 2mm
Monoclinique	Monoclinique	Monoclinique	13	P, C	2/m, m, 2
Triclinique	Triclinique	Triclinique	2	P	-1, 1

Les groupes d'espace en trois dimensions sont formés par des combinaisons des 32 groupes ponctuels cristallographiques avec les 14 réseaux de Bravais, chacun de ces derniers appartenant à l'un des 7 systèmes de réseau. Cela signifie que l'action de tout élément d'un groupe d'espace donné peut être exprimée comme l'action d'un élément du groupe ponctuel approprié, suivie éventuellement d'une translation.

Un groupe d'espace est donc une combinaison de la symétrie translationnelle d'une maille élémentaire (y compris le centrage du réseau), des opérations de symétrie du groupe ponctuel (réflexion, rotation et rotation impropre, également appelée roto-inversion), ainsi que des opérations de symétrie d'axe hélicoïdal et de plan de glissement.

Les groupes d'espace (groupes de symétrie d'une configuration dans l'espace) sont classés en systèmes cristallins selon leurs groupes ponctuels, et en systèmes de réseau selon leurs réseaux de Bravais. Les systèmes cristallins qui ont des groupes d'espace attribués à un système de réseau commun sont combinés en une famille cristalline.

Systèmes Cristallins, Exigences et Intuitions

Système Cristallin	Symétrie Caractéristique	Exigences	Intuitions
Isométrique	4 axes ternaires de rotation	$a = b = c, \alpha = \beta = \gamma = 90^\circ$	Symétrie maximale : cubes, octaèdres.

Système Cristallin	Symétrie Caractéristique	Exigences	Intuitions
Tétragonal	1 axe quaternaire de rotation	$a = b, \alpha = \beta = \gamma = 90^\circ$	Comme le cubique, mais avec un axe allongé (c).
Hexagonal	1 axe sennaire de rotation	$a = b, \alpha = \beta = 90^\circ, \gamma = 120^\circ$	Symétrie élevée avec un axe principal (ex. : prismes hexagonaux).
Trigonal	1 axe ternaire de rotation	$a = b, \alpha = \beta = 90^\circ, \gamma = 120^\circ$	Similaire à l'hexagonal, mais avec une symétrie ternaire (ex. : quartz).
Orthorhombique	3 axes binaires de rotation	$\alpha = \beta = \gamma = 90^\circ$	Étendre la logique monoclinique à 3 axes (ex. : prismes orthorhombiques).
Monoclinique	1 axe binaire de rotation	$\alpha = \gamma = 90^\circ$	Rotation de 180° autour de l'axe c, sans créer de vides dans le plan de base.
Triclinique	Aucune, ou seulement un centre de symétrie	aucune	Symétrie minimale : pas de rotation, seulement une inversion possible.

Systèmes Cristallins, Classes Cristallines, Formes et Symétries

Système Cristallin	Classe Cristalline	Hermann-Mauguin	Symétries	Formes Courantes	Groupe Abstrait
Isométrique	Hexoctaédrique	4/m -3 2/m	3 A_4 , 4 A_3 , 6 A_2 , 9 M, C	Cube {100}, octaèdre {111}, dodécaèdre {110}	$S_4 \times Z_2$
Isométrique	Diploïdale	2/m -3	3 A_2 , 4 A_3 , C	Cube {100}, diploïde {321}	$A_4 \times Z_2$
Tetragonale	Ditetragonale dipyramidale	4/m 2/m 2/m	1 A_4 , 4 A_2 , 5 P, C	Prismes, dipyramides	$D_8 \times Z_2$
Hexagonal	Dihexagonale dipyramidale	6/m 2/m m	1 A_6 , 6 A_2 , 6 M, C	Prisme hexagonal {10-10}, bipyramide hexagonale {11-21}	$D_{12} \times Z_2$
Trigonal	Rhomboédrique	-3m	1 A_3 , 3 A_2 , 3 M, C	Rhomboèdre {10-11}, scalénoèdre {21-31}	Cyclique $Z_6 = Z_2 \times Z_3$
Trigonal	Trapézoédrique	3 2	1 A_3 , 3 A_2	Prisme trigonal {10-10}, rhomboèdre {10-11}	diédrale $D_6 = D_2 \times D_3$

Système Cristallin	Classe Cristalline	Hermann-Mauguin	Symétries	Formes Courantes	Groupe Abstrait
Orthorhombique	Dipyramide orthorhombique	2/m 2/m 2/m	3 A ₂ , 3 P, C	Pinacoïdes : {100}, {010}, {001} ; prismes : {100}, {010}, {001}, dipyramide {111}	Klein 4 V = Z ₂ x Z ₂
Monoclinique	Prismatique	2/m	1 A ₂ , 1 P, C	Pinacoïdes : {100}, {010}, {001} ; prismes : {0kl}, {hk0}	Klein 4 V = Z ₂ x Z ₂
Triclinique	Pinacoïdale	1	C	Pinacoïdes : {h00}, {0k0}, {00l}, {h0l}, {0kl}, {hk0}, {hkl} (2 faces chacun)	Cyclique Z ₂

20 Minéraux de Référence

Minéral	Système Cristallin	Classe Cristalline	Hermann-Mauguin	Formes Typiques	Plans de Clivage	Qualité de Clivage	Groupe d'Espace	Formule Chimique
Diamant	Isométrique	Hexoctaédrique	4/m -3 2/m	Octaèdre {111}, cube {100}, dodécaèdre {110}	{111}	parfait	Fd3m	C
Fluorite	Isométrique	Hexoctaédrique	4/m -3 2/m	Cube {100}, octaèdre {111}, dodécaèdre {110}	{111}	parfait	Fm3m	CaF ₂
Pyrite	Isométrique	Diploïdale	2/m -3	Cube {100}, diploïde {321}	{001}	imparfait	Pa3	FeS ₂
Almandin (Grenat)	Isométrique	Hexoctaédrique	4/m -3 2/m	Dodécaèdre rhomboïdal {110}	aucun	N/A	Ia3d	Fe ₃ Al ₂ (SiO ₄) ₃
Apatite	Hexagonal	Dipyramidale	6/m	Prismes hexagonaux, aiguilles	{0001}	mauvais	P6 ₃ /m	Ca ₅ (PO ₄) ₃ (OH, F, Cl)
Quartz (cristal de	Trigonal	Trapézoédrale	32	Prismes hexagonaux, 3	{1011}	mauvais	P3 ₁ 21	SiO ₂

Minéral	Système Cristallin	Classe Cristalline	Hermann-Mauguin	Formes Typiques	Plans de Clivage	Qualité de Clivage	Groupe d'Espace	Formule Chimique
roche)				2bipyramides trigonales				
Calcite	Trigonal	Hexagonale scalénoédrale	-3 2/m	Rhomboèdre {10-11}, scalénoèdre {21-31}	{1011}	parfait	R-3c	CaCO ₃
Corindon	Trigonal	Hexagonale scalénoédrale	-3 2/m	Dodécaèdre rhomboïdal, barils	aucun	N/A	R-3c	Al ₂ O ₃
Leucite	Tétragonal	Dipyramidale	4/m	Trapézoèdres	{110}	mauvais	I4 ₁ /a	KAlSi ₂ O ₆
Vésuvianite	Tétragonal	Bipyramidale tétragonale	4/m 2/m 2/m)	Prismes courts, pyramides tétraédriques	{110}	mauvais	P4/nnc	Ca ₁₀ (Mg,Fe) ₂ Al ₄ (SiO ₄) ₅ (Si ₂ O ₇) ₂ (OH) ₄
Topaze	Orthorhombique	Dipyramidale	2/m 2/m 2/m	Prismes, pyramides	(001)	parfait	Pnma	Al ₂ SiO ₄ (F,OH) ₂
Aragonite	Orthorhombique	Dipyramidale	2/m 2/m 2/m	Prismes, aiguilles	{010}	distinct	Pmcn	CaCO ₃
Orthose	Monoclinique	Prismatique	2/m	Tablettes, macles de Karlsbad	{001}, {010}	Parfait/bon	B2/m	KAlSi ₃ O ₈
Gypse	Monoclinique	Prismatique	2/m	Prismes, masses fibreuses, tablettes	{010}, ...	parfait	P2 ₁ /c	CaSO ₄ ·2H ₂ O
Heulandite	Monoclinique	Prismatique	2/m	Tablettes, prismes	{010}	parfait	C2/c	(Na,Ca) ₄ Al ₆ Si ₁₀ O ₃₂ ·12H ₂ O
Talc	Triclinique	Pinacoïdale	1	Feuillets, masses micacées	{001}	parfait	P-1	Mg ₃ Si ₄ O ₁₀ (OH) ₂